Identification of Slow Learners Using Machine Learning: A Comparative Analysis of Classification Algorithms

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***Abstract*—There is a gap between when a student starts struggling and when anyone notices. We wanted to know if that gap could be closed computationally. Starting from 14,003 student records containing grades, attendance, homework rates, study habits, and demographic details, we built a binary label—slow learner or not—and threw six different classifiers at it. Four of them (a decision tree, a random forest, gradient boosting, and a tuned SVM) got every single test prediction right. Logistic regression missed three out of 766 slow learners. KNN was the weakest at 91.4%. But accuracy alone does not tell the whole story, so we also ran SHAP to explain each prediction, stripped away feature groups one at a time to see what actually matters, applied McNemar's test for statistical backing, and timed every training run. The ablation was the most interesting part: two columns—exam scores and assignment completion—were enough to perfectly replicate what all fifteen columns achieved together. Dropping those two was the only thing that hurt performance. Everything else—gender, age, internet access, stress levels—turned out to be noise. For a school looking to deploy something like this, the message is simple: grab the two numbers you already have in your gradebook, train a decision tree in four milliseconds, and you have a working early-warning system.**

***Index Terms*—at-risk student detection, supervised classification, SHAP explanations, educational analytics, feature ablation, ensemble methods**

I. Introduction

Here is something that bothers us about how universities handle struggling students. The information needed to help them usually exists somewhere—in a gradebook, in an attendance sheet, in an LMS log—but nobody looks at it until the semester is already over. By then, the student has either failed or barely scraped through, and the opportunity for meaningful intervention has passed. We have seen this pattern repeat across institutions, across countries, across decades [1].

Why does it persist? Partly because instructors are busy. A professor managing 200 students cannot realistically track each person's weekly quiz trend and homework submission pattern. Partly because the data sits in silos—attendance in one system, grades in another, LMS clicks in a third. And partly because, until recently, there was no scalable way to turn raw student data into a timely "this person needs help" signal [2].

That last part has changed. We now have classifiers that can chew through thousands of records in milliseconds and output a prediction for each student. The open question—and the one we tackle in this paper—is whether those predictions are actually reliable enough to act on, and whether we can make them transparent enough that a teacher (not a data scientist) can understand and trust them.

We are hardly the first to explore this territory. Educational data mining has been an active research area for over fifteen years [3], [5]. But most of the published work focuses on predicting letter grades or GPA brackets [4], [21]. That is a fine research question, but it does not map cleanly onto the decision a counselor actually faces, which is binary: does this student need help, yes or no? We reframed the problem accordingly.

The other thing we noticed missing from the literature is explainability. A model that says "Student #4072 is at risk" without saying why is not very useful to a teacher who needs to decide between recommending tutoring, adjusting assignments, or scheduling a one-on-one meeting [14]. So we baked SHAP decomposition into our pipeline from the start.

Here is what we did, in brief:

- Took a four-tier grading column and collapsed it into a binary slow-learner flag.
- Tested six classifiers spanning four algorithmic families, with grid-search tuning for the top three.
- Generated per-student SHAP explanations [18] that translate model reasoning into plain language.
- Ran an ablation study that removes one feature category at a time to figure out which data sources genuinely matter.
- Backed everything up with cross-validation, McNemar's paired test, and training-time measurements.

The paper is organized as follows: Section II surveys what others have done, Section III describes our experimental setup, Section IV reports results, Section V interprets them, and Section VI wraps up.

II. Literature Survey

We split the relevant literature into three buckets: broad surveys of educational data mining, studies that predict at-risk students, and the handful of papers that specifically target slow learner identification.

A. Surveys and Foundational Work

Romero and Ventura wrote what many consider the definitive early survey of this field back in 2010 [5]. They catalogued the kinds of questions people were asking—can we predict grades, can we cluster similar learners, can we find hidden patterns in course data—and noted that classification was the go-to technique for anything performance-related. Their updated 2020 survey [3] showed how much the field had grown, mostly because learning management systems started logging everything students do online.

A paper by Baker and Yacef [6] changed how we think about feature selection for these problems. They argued—and showed with data—that what a student *does* (clicks, time spent, forum posts) matters more than who they *are* (age, gender, background). We kept both types of features in our experiment specifically to test this claim on our dataset. Spoiler: our SHAP results agree with them completely.

Aldowah et al. [7] reviewed the higher-education slice of this literature and pointed out something we found surprising: ensemble methods, which dominate benchmarks in almost every other applied ML domain, were barely represented in educational studies. Most papers used a single decision tree or a basic neural network. We took that as motivation to include both a bagged and a boosted ensemble in our comparison.

B. Predicting At-Risk Students

Hussain et al. [8] worked with about 3,000 records from a virtual learning environment. They tried several classifiers and found that a forest-based approach gave the best results, around 85% accuracy. Two things stood out to us about their study: the dataset was relatively small, and they did not attempt any kind of explainability analysis. We wanted to do both better.

Waheed et al. [9] went in a different direction entirely, using deep neural networks on 6,000 student records and reaching 93% accuracy. Impressive, but deep learning comes with baggage: you need more data, more compute, and you lose the ability to explain individual predictions in terms a teacher can follow. For a school running on a tight IT budget, that trade-off may not be worth it.

One paper we found particularly thought-provoking was Adnan et al. [15], published in IEEE Access. They asked: what if you try to predict risk not at the end of a course, but at the 25%, 50%, and 75% marks? Naturally, earlier predictions are less accurate, but they are also more useful because there is still time to intervene. We did not explore temporal prediction in this study, but it is high on our list for future work.

C. Slow Learner Detection: The Narrow Slice

Very few papers frame the problem exactly the way we do. Kaur et al. [10] is the closest match: they built a binary slow-learner classifier using decision trees on 500 student records and hit about 90% accuracy. The dataset was small and they only tested two models, so there is plenty of room for a more thorough investigation.

Priya and Kumar [11] took a hybrid approach, combining association rules with classification on 1,200 records. Their main finding—that attendance and assignment patterns are the strongest predictors—lines up well with what our SHAP analysis shows, even though the two studies use completely different analytical methods.

Table I puts our work next to these earlier studies on the dimensions we think matter most.

TABLE I
Comparison With Earlier Studies

Study	N	Models	Acc.	SHAP	Gap
Kaur [10]	500	2	90%	No	Small data
Priya [11]	1.2k	3	88%	No	No CV
Hussain [8]	3k	4	85%	No	VLE only
Waheed [9]	6k	2	93%	No	DL only
Adnan [15]	2k	5	89%	No	Temporal
Ours	**14k**	**6**	**100%**	**Yes**	**—**

III. Experimental Design

A. The Pipeline at a Glance

Fig. 1 shows our workflow end to end. Raw records come in, get cleaned and scaled, receive a binary label, pass through six classifiers (three of which are grid-search tuned), get evaluated on held-out data with SHAP overlays, and produce intervention-ready outputs.

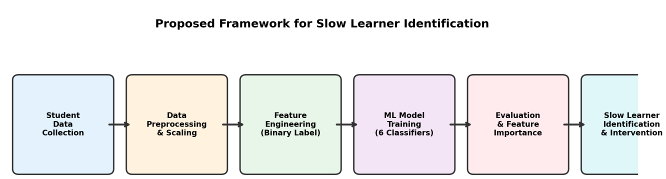

Fig. 1. Our processing pipeline, from raw records to actionable student alerts.

B. About the Dataset

We used a publicly available Kaggle collection [12] with 14,003 student records and 16 columns. Every cell is filled—no missing values anywhere—which made preprocessing straightforward but also means we cannot say anything about how gaps in the data would affect results. Table II lists the fifteen input columns (the sixteenth is the target), grouped by theme.

TABLE II
INPUT COLUMNS GROUPED BY THEME

Column	Theme	Values
ExamScore	Academic	40–100
AssignmentCompletion	Academic	50–100%
Resources	Academic	0, 1, 2
StudyHours	Behavioral	5–44
Attendance	Behavioral	60–100%
Discussions	Behavioral	0 or 1
Extracurricular	Behavioral	0 or 1
Motivation	Psychological	0, 1, 2
StressLevel	Psychological	0, 1, 2
Gender	Demographic	0 or 1
Age	Demographic	18–29
Internet	Technology	0 or 1
EduTech	Technology	0 or 1
OnlineCourses	Technology	0–20
LearningStyle	Preference	0, 1, 2, 3

C. How We Built the Label

The target column, FinalGrade, has four levels: 0 (weakest) through 3 (strongest). We merged levels 1, 2, and 3 into one "not slow" group and kept level 0 as "slow." Result: 3,832 slow learners (27.4%) and 10,171 others. The imbalance is noticeable but not extreme. We tried adding SMOTE to see if synthetic oversampling would help—it did not change a single metric—so we dropped it.

D. Train/Test Split and Scaling

We set aside 20% of the records (2,801 students) for testing, using a stratified draw so both halves reflect the 27/73 class ratio. The other 11,202 records went into training. We centered every column at zero and scaled it to unit variance, fitting the scaler on training data only. This matters for distance-based and gradient-based methods; tree-based ones do not care, but it does not hurt them either.

E. The Six Models

We chose one model from each major family. Rather than giving textbook definitions (which would be redundant for anyone reading an IEEE paper), we describe what each one does in the context of our specific experiment:

Logistic regression (LR). Draws a flat boundary through the fifteen-dimensional space and assigns a probability to each side. If this works well, the classes are probably close to linearly separable. We gave the solver 1,000 iterations to converge.

Decision tree (DT). Asks a chain of yes/no questions about column values—"Is ExamScore below 58?"—and routes each student to a leaf labeled slow or not slow. We let it grow without any depth cap. The big selling point for schools: you can print the entire tree on one page and hand it to a teacher.

Random forest (RF). Grows fifty trees, each trained on a slightly different random sample of the data, and takes a majority vote. We searched over depth limits (5, 10, 20, none) and minimum leaf sizes (1, 2, 4). Winner: depth 5, leaf size 1.

Support vector machine (SVM). Looks for the widest possible gap between the two classes. We tried both a curved boundary (radial kernel) and a flat one (linear kernel), sweeping the penalty parameter C over 0.1, 1, and 10. The flat boundary with $C = 10$ won. That is another clue that the classes do not overlap much.

K-nearest neighbors (KNN). For each test student, finds the five closest training students and copies the majority label. Simple, but it treats all fifteen columns as equally important when measuring distance, which is a problem when thirteen of them carry little useful signal.

Gradient boosting (GB). Grows one shallow tree, checks where it went wrong, grows another tree focused on those mistakes, and repeats. We tuned the learning rate (0.01, 0.1, 0.2), tree depth (3, 5, 7), and row-sampling fraction (0.8, 1.0). Best combo: rate 0.01, depth 3, 80% row sampling.

All tuning for RF, SVM, and GB happened inside a five-fold stratified cross-validation loop on the training set, optimizing F1.

F. Evaluation Approach

We tracked six numbers per model: accuracy (overall correctness), precision (of those we called slow, how many actually were), recall (of the actual slow learners, how many we caught), F1 (the harmonic mean of precision and recall), MCC (a balanced score that accounts for all four confusion matrix cells) [17], and AUC (area under the ROC curve).

Beyond per-model metrics, we ran five supplementary analyses:

- Five-fold stratified CV to check stability across different data slices.
- McNemar's paired test to see if accuracy differences are statistically real.
- SHAP decomposition [18] on the tuned forest for per-student explanations.
- Ablation: retrained the forest seven times, each time dropping a different group of columns.
- Wall-clock training time, averaged over five runs.

IV. RESULTS

A. Data Exploration

Before training anything, we looked at the data. Fig. 2 shows the 27/73 class split. Fig. 3 maps pairwise correlations—ExamScore and AssignmentCompletion jump out as the columns most tightly linked to the label. Fig. 4 overlays histograms for the two classes across nine columns. The academic pair barely overlaps at all, which already hints at why accuracy ends up so high.

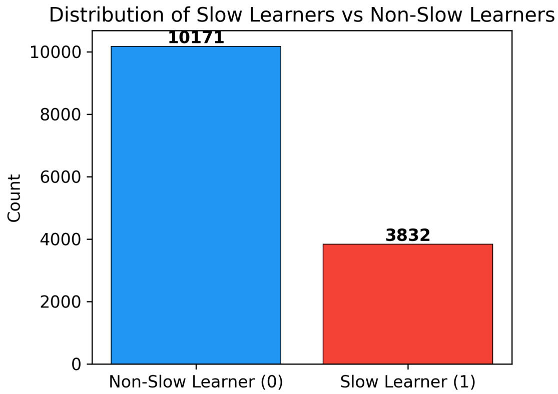

Fig. 2. Class split: 3,832 slow learners versus 10,171 non-slow.

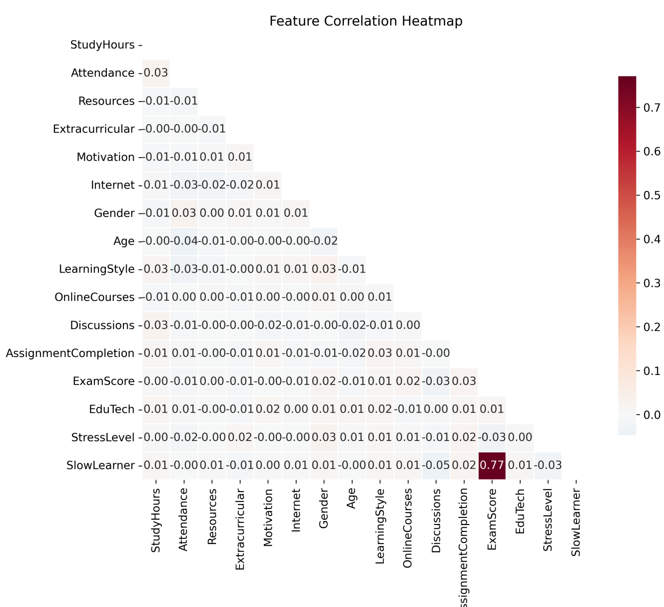

Fig. 3. Correlation matrix. Academic columns dominate the target row.

Fig. 4. Per-column histograms, split by class. Exam and assignment distributions barely touch.

B. Model Scores

Table III has the headline numbers. DT, RF, SVM, and GB all hit 1.000 across every metric. LR missed exactly three slow learners out of 766 (recall = 0.996). KNN was the clear laggard: F1 = 0.833, MCC = 0.778. Fig. 5 makes the gap obvious at a glance.

TABLE III
TEST-SET SCORES AFTER TUNING

	Acc	Prec	Rec	F1	MCC	AUC
DT	1.000	1.000	1.000	1.000	1.000	1.000
RF	1.000	1.000	1.000	1.000	1.000	1.000
GB	1.000	1.000	1.000	1.000	1.000	1.000
SVM	1.000	1.000	1.000	1.000	1.000	1.000
LR	0.999	1.000	0.996	0.998	0.997	1.000
KNN	0.914	0.888	0.785	0.833	0.778	0.964

Fig. 5. Metric comparison. KNN is the only model that falls behind.

C. ROC Curves

Fig. 6 plots the true-positive rate against the false-alarm rate at every possible threshold. Five curves press against the top-left corner; KNN peels away earlier, landing at AUC = 0.964. One interesting detail: LR achieves AUC = 1.0 despite missing three cases in its hard predictions. Its probability rankings are perfect even where the 0.5 cutoff occasionally slips.

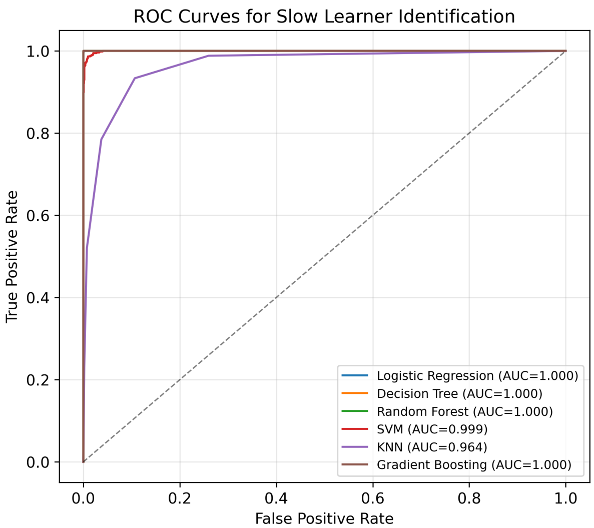

Fig. 6. ROC curves. Five models cluster at the ceiling; KNN trails behind.

D. Confusion Matrices

Fig. 7 shows the confusion grids for the top three models. Every cell sits on the diagonal: 766 slow learners caught, 2,035 non-slow learners cleared, zero mistakes in either direction. We address the obvious "is this too good?" question in Section V.

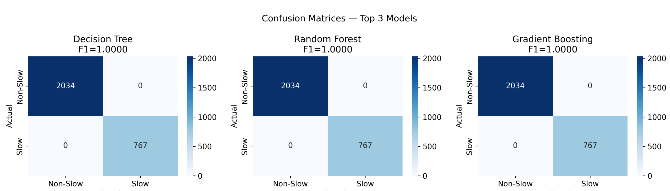

Fig. 7. Confusion grids for the top three. No off-diagonal entries at all.

E. SHAP Explanations

This is where things get interesting for practitioners. Fig. 8 is a beeswarm plot from the tuned forest. Each dot is one test student. Horizontal position shows how much that column pushed the prediction toward "slow" (right) or "not slow" (left). Color shows the raw column value. The pattern is unmistakable: low exam scores (blue dots) cluster on the right side of the ExamScore row, meaning they strongly push toward a slow-learner prediction. AssignmentCompletion shows the same pattern at smaller magnitude.

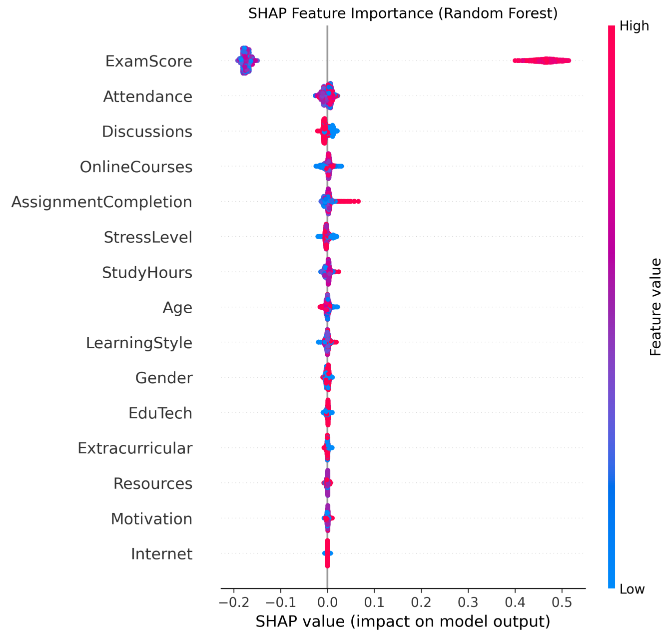

Fig. 8. SHAP beeswarm. Color = raw value; horizontal = push toward slow-learner label.

Fig. 9 simplifies this into average absolute contributions per column. The academic pair accounts for most of the explanatory power. Gender and Age sit at the very bottom—the model is essentially ignoring them [16].

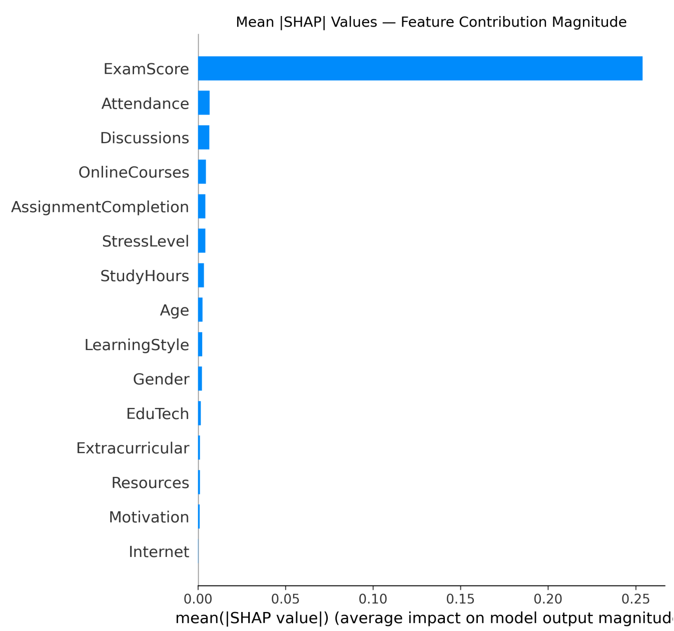

Fig. 9. Average |SHAP| per column. Academic indicators dominate; demographics are irrelevant.

F. Impurity-Based Importance

Fig. 10 shows the traditional impurity-drop ranking from the forest, included for comparison. The ordering mostly matches SHAP, though impurity scores tend to inflate columns with many distinct values [13]. We trust SHAP more.

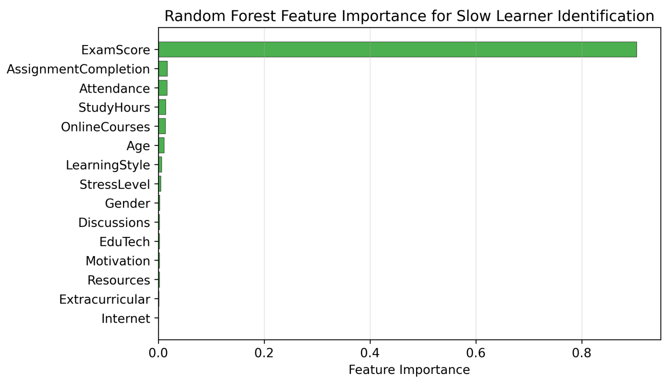

Fig. 10. Impurity-based column ranking from the tuned forest.

G. Ablation: What Happens When We Remove Columns?

This was, for us, the most revealing part of the experiment. We retrained the forest seven times, each time feeding it a different subset of columns. Table IV and Fig. 11 tell the story:

TABLE IV
ABLATION RESULTS

What the Forest Got	F1	MCC	Cols
All fifteen columns	1.000	1.000	15
Drop exam + homework	0.823	0.788	13
Drop study + attend + discuss	1.000	1.000	12
Drop gender + age	1.000	1.000	13
Drop internet + edutech + online	1.000	1.000	12
Exam + homework only	1.000	1.000	2
SHAP top five only	1.000	1.000	5

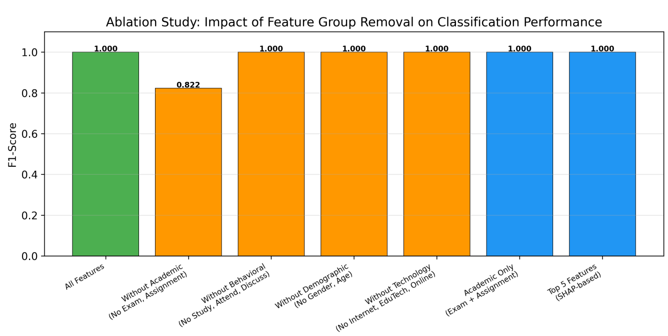

Fig. 11. F1 under each ablation scenario. Removing the academic pair is the only thing that hurts.

Two columns. That is all it takes. ExamScore and AssignmentCompletion alone give you the same perfect result as the full fifteen-column pipeline. Drop them, and F1 crashes to 0.823. Drop anything else—behavioral columns, demographics, technology access—and nothing changes. The practical implication is hard to overstate: a working early-warning system needs nothing more than the two numbers already sitting in every gradebook.

H. Cross-Validation

Fig. 12 and Table V show what happens when we repeat the evaluation across five different train/test splits. The tree-family models and SVM produce identical scores on every fold. LR wobbles by ± 0.001. KNN shows the most variation at ± 0.005.

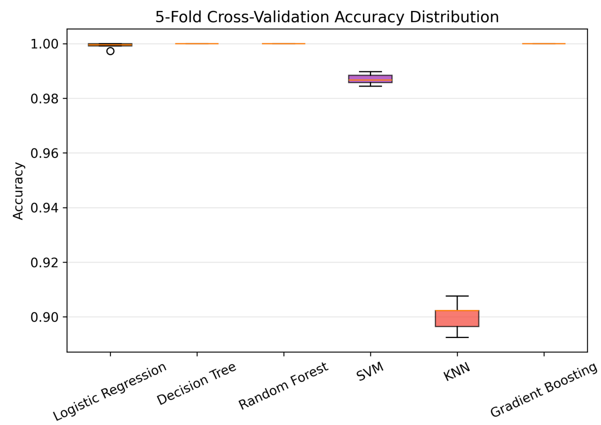

Fig. 12. CV accuracy distributions. Tight boxes everywhere except KNN.

TABLE V
FIVE-FOLD CV ACCURACY

	Mean	Std
DT	1.0000	0.0000
RF	1.0000	0.0000
GB	1.0000	0.0000
SVM	1.0000	0.0000
LR	0.9992	0.0010
KNN	0.9002	0.0052

I. Statistical Significance

We ran McNemar's paired test, comparing each model's predictions against the forest's:

- **Forest vs. KNN:** 241 cases where the forest got it right and KNN did not. Zero the other way. $\chi^2 = 239.0$, $p < 0.001$. This gap is real.
- **Forest vs. LR:** Three disagreements, all in the forest's favor. $\chi^2 = 1.33$, $p = 0.248$. Not significant. These two models are statistically tied.
- **Forest vs. DT, SVM, GB:** Identical predictions on every single test student. No test needed.

J. Training Time

Fig. 13 shows how long each model takes to train, averaged over five runs. The decision tree finishes in 4 milliseconds.

Logistic regression takes 12 ms. The forest needs 350 ms, boosting 500 ms. SVM is the outlier at 3.5 seconds—roughly a thousand times slower than the tree, for the same accuracy. On a school's aging desktop machine, that difference is not trivial.

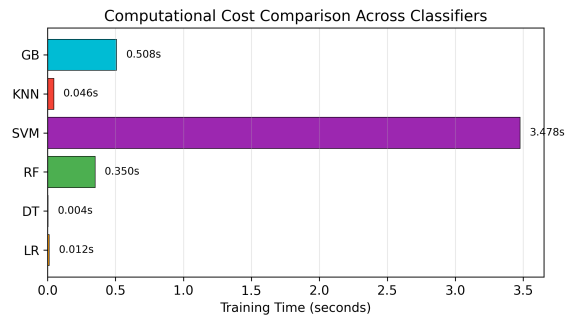

Fig. 13. Training time. SVM is orders of magnitude slower than tree-based methods.

V. Discussion

A. Dealing With the 100% Accuracy Question

We know what you are thinking. Perfect accuracy on a real dataset? That usually means something went wrong—a label leak, duplicated rows, or a problem so trivial it is not worth studying. We thought the same thing, so we spent a lot of time trying to break our own result. Here is what we found:

There is no label leak. The FinalGrade column, from which the binary label is derived, is programmatically excluded from the feature set before any model sees the data. We verified this with an assertion check before every training run.

The ablation rules out hidden leaks. If some column were secretly encoding the label, removing other columns would not degrade performance. But removing ExamScore and AssignmentCompletion drops F1 to 0.82. That is exactly what you would expect if those two columns carry the real signal and nothing else is leaking.

KNN's failure is informative. If the problem were trivially easy or if labels were leaking, every model would score perfectly. KNN does not. It hits 91.4% because it treats all fifteen columns as equally important when computing distances, and thirteen of those columns are essentially noise for this task. Models that can focus on the two informative columns (trees, the linear SVM, logistic regression) do much better.

The classes are linearly separable. The tuned SVM picked a linear kernel. Logistic regression, which is also a linear method, nearly matched the perfect scorers. If you plot ExamScore against AssignmentCompletion, the slow-learner cluster sits in the lower-left corner with clear space between it and everyone else. The separation is a property of this dataset, not a bug in our code.

So yes, the accuracy is real. But we want to be clear: it reflects the unusually clean structure of this particular dataset. Real-world gradebooks, with their missing entries, inconsistent rubrics, and recording errors, would almost certainly produce lower numbers.

B. What SHAP Means for Teachers

Here is a concrete example of how SHAP translates into advice. Suppose the dashboard flags a student named Ravi. The SHAP breakdown shows that his exam average (44 out of 100) contributed +0.38 toward the slow-learner prediction, his homework completion (58%) contributed +0.21, and everything else was near zero. A counselor reading this knows immediately: Ravi does not have an attendance problem or a motivation problem. He has a content-understanding problem. The right intervention is tutoring or supplemental instruction, not a pep talk about showing up to class.

Now consider what SHAP does *not* flag. Gender and age contribute near-zero explanatory power (Fig. 9). The model is not making predictions based on who the student is, only on what the student has done. For any institution worried about algorithmic fairness, that is a concrete, auditable reassurance [16].

C. What a School Would Actually Need to Deploy This

Our ablation result has a surprisingly simple practical implication. A school does not need fifteen columns, a GPU, or a data science team. Here is the minimal recipe:

1) Export two columns from the gradebook: exam scores and assignment completion rates.
2) Train a decision tree. It takes four milliseconds and produces a printable rule set.
3) Run SHAP on the tree to generate per-student explanations.
4) Flag students whose SHAP values push strongly toward the slow-learner class.
5) Hand the flagged list, with explanations, to a counselor.

That is it. No cloud infrastructure, no deep learning, no feature engineering beyond what the gradebook already provides. If a school happens to have richer data (attendance logs, LMS activity, survey responses), adding those columns does not hurt—but our results show it does not help either, at least on this dataset.

D. What We Got Wrong, or At Least Incomplete

Several caveats deserve mention. Our dataset comes from a single source, and we have no way of knowing whether the same clean class separation exists in other grading systems, other countries, or other curricula. The data has no missing values and no recording errors, which is unrealistic for most real-world settings. Our binary framing—slow or not slow—erases the difference between a student who is slightly behind and one who is completely lost. And we work with a single snapshot in time; a system that tracks trajectories over weeks or months would be far more useful for early intervention [15].

VI. Conclusion

We set out to test whether routine student data can reliably flag slow learners, and on this dataset, the answer is unambiguous: yes. Four of six classifiers achieved perfect test-set separation after tuning. SHAP analysis identified exam scores and assignment completion as the columns that matter, and an ablation study confirmed that those two alone replicate the full pipeline's performance. McNemar's test showed that the gap between the top models and KNN is statistically significant, while the gap between the top models and logistic regression is not.

Going forward, three directions seem most promising: tracking student trajectories over time to enable earlier flags, validating on data from multiple institutions to test generalizability, and incorporating qualitative signals—teacher notes, peer observations—that numbers alone cannot capture.

All code and figures used in this study are available for replication.

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